Evaluation of Algorithms for Fractal Dimension Estimation in Medical Images: Complexity and Performance Analysis

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Resumo

Resumo do artigo em portugês

Palavras-chave: Palavras chave em português

Resumo

Abstract
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1 Introduction

Fractal geometry, formalised by Benoit B. Mandelbrot in his landmark monograph *The Fractal Geometry of Nature* (MANDELBROT, 1982), provided a mathematical framework for describing the roughness, branching, and statistical self-similarity pervasive in natural phenomena that classical Euclidean geometry fails to capture. The central quantitative descriptor of this framework is the fractal dimension D, estimated in practice as the negative slope of a linear regression over a log-log plot of a covering measure against the measurement scale ε. Shortly after Mandelbrot's unification, Pentland (PENTLAND, 1984) demonstrated that fractal dimension could be computed directly from digital imagery to discriminate surface textures in natural scenes — a result that bridged fractal geometry and image analysis, and opened the way for its systematic application to medical imaging tasks.

The adoption of fractal analysis in medical imaging occurs since the 1990s, alongside the proliferation of high-resolution digital modalities — multi-detector computed tomography, 3 T magnetic resonance imaging, and digital mammography. Applications spanning pulmonary emphysema (MISHIMA *et al.*, 1999), retinal vascular morphology (JELINEK; FERNANDEZ, 2000), brain cortical folding (FREE *et al.*, 1996), and tumour margin irregularity (RANGAYYAN; NGUYEN, 2007) demonstrated that D encodes structural properties that classical first-order statistical descriptors — variance, entropy, autocorrelation — could not adequately capture. From the 2010s onward, fractal descriptors were incorporated into hybrid feature engineering pipelines alongside learned representations (LOPES; BETROUNI, 2011; KING *et al.*, 2010), and more recently, fractal dimension has been explored as a a feature describer to enhance the interpretability of deep learning models (ROBERTO *et al.*, 2021a) and as a data augmentation strategy with GANs (BORGUE *et al.*, 2025a). Contemporary clinical datasets — commonly comprising images at resolutions exceeding 1024×1024 pixels — elevated the algorithmic efficiency of dimension estimation from a secondary concern to a primary design constraint. The central question shifted from whether fractal dimension is discriminative to at what computational cost it can be reliably extracted, and which algorithm offers the optimal balance of precision, robustness, and efficiency under realistic operating conditions.

Addressing this question requires a rigorous algorithmic engineering perspective. The four methods selected for analysis span the principal design axes of the problem: binary versus grayscale input, and non-overlapping versus dense-overlap covering strategies. The *Classical Box-Counting* (BC) (MANDELBROT, 1982) partitions the binary image into a grid of non-overlapping boxes of decreasing side and counts how many contain foreground pixels, yielding a coarse but efficient estimate of D. The *Gliding Box* (GB) (ALLAIN; CLOITRE, 1991) extends this idea by sliding a window across every pixel position, producing a richer mass-frequency distribution at the cost of processing a far larger number of box placements. The *Differential Box-Counting* (DBC) (SARKAR; CHAUDHURI, 1994) lifts the problem to grayscale images by treating the intensity surface as a three-dimensional relief and counting the column of boxes required to span the local intensity range within each cell, thereby retaining gradient information that binary methods discard. The *Blanket Method* (BM) (PELEG *et al.*, 1984) estimates the fractal dimension of the intensity surface by measuring how its apparent area grows as a morphological cover is progressively inflated and deflated around it at each scale. Contrasting these four methods also surfaces a fundamental trade-off inherent to the preprocessing stage: binarisation lowers the representational cost of the input, but discards intensity gradients that carry clinically significant discriminatory information — a loss that the grayscale methods avoid at the price of higher computational demand.

This work addresses the following research objectives:

- **Algorithmic analysis:** for each of the four methods, in both naive and structurally optimised formulations, derive formal asymptotic bounds for time and auxiliary space complexity and validate them against empirical measurements on the UCSB medical image dataset, contrasting theoretical scaling predictions with observed wall-clock behaviour.
- **Pre-processing and implementation cost:** quantify the overhead imposed by mandatory pre-processing stages and characterise the data-structure choices that drive the efficiency gap between naive and optimised implementations.
- **Effectiveness and trade-off synthesis:** assess the discriminatory power of each algorithm by measuring the statistical separation it induces between pathological and healthy tissue classes in terms of the estimated fractal dimension, and synthesise all time, memory, and effectiveness evidence into a structured trade-off comparison across the binary-versus-grayscale and naive-versus-optimised design axes.

The remainder of this article is structured as follows: Section 2 establishes the mathematical foundations of fractal analysis and formally defines each algorithm; Section 3 presents pseudocode and derives asymptotic complexity bounds; Section 4 describes the experimental methodology, dataset, and evaluation metrics; Section 5 reports and discusses empirical results; Section 6 concludes with recommendations and directions for future work.

2 Theoretical Foundation

2.1 Fractal Analysis and Dimension Estimation

Fractal geometry provides a mathematical framework for characterising structures whose complexity persists across observation scales, a property termed *statistical self-similarity* (MANDELBROT, 1982). Fractal sets exhibit power-law scaling: a characteristic measure $M(\varepsilon)$ obtained at scale ε satisfies

$$M(\varepsilon) \sim C \cdot \varepsilon^{-D}, \tag{1}$$

where C is a proportionality constant and D is the fractal dimension (FALCONER, 2003). Taking logarithms of both sides yields the linear relationship

$$\log M(\varepsilon) = \log C - D \log \varepsilon, \tag{2}$$

from which D is recovered as the negative slope of a least-squares regression over the log-scale pairs $\{(\log \varepsilon_i, \log M(\varepsilon_i))\}_{i=1}^{m}$, computed across m discrete scales.

Because the Hausdorff dimension (HAUSDORFF, 1919) requires an infimum over all possible coverings, it is not directly computable on discrete pixel grids. The Minkowski–Bouligand (box-counting) dimension D_B provides a tractable approximation (FALCONER, 2003):

$$D_B = \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}, \tag{3}$$

where $N(\varepsilon)$ is the minimum number of boxes of side ε needed to cover the set. In practice, Eq. (3) is approximated by the finite regression of Eq. (2) over a scale set $\mathcal{S} = \{\varepsilon_1 > \varepsilon_2 > \cdots > \varepsilon_m\}$. The quality of the estimate is assessed through the coefficient of determination R^2 of the regression: values close to unity confirm consistent power-law scaling and are a necessary condition for interpreting D_B as a meaningful fractal descriptor (LOPES; BETROUNI, 2011).

2.2 Classical Box-Counting Method

The Classical Box-Counting (BC) method is the direct discretisation of Eq. (3) for binary images (MANDELBROT, 1982; LIEBOVITCH; TOTH, 1989). Let I be a binary image of $N = W \times H$ pixels. For a given box side ε, the image is partitioned into $\lceil W/\varepsilon \rceil \times \lceil H/\varepsilon \rceil$ non-overlapping cells; a cell is *occupied* if it contains at least one foreground pixel:

$$N(\varepsilon) = \sum_{i=0}^{\lceil W/\varepsilon \rceil - 1} \sum_{j=0}^{\lceil H/\varepsilon \rceil - 1} \mathbf{1}[\exists\,(x,y) \in B_{ij}(\varepsilon) : I(x,y) = 1]\,. \tag{4}$$

Because BC operates on binary images, a prior binarisation step is required when the input is a grayscale medical image. The standard strategy is Otsu's global thresholding (OTSU, 1979), which selects the threshold t^* that minimises intra-class intensity variance. This step introduces an inherent trade-off: binarisation lowers the computational cost of the subsequent coverage test, but irreversibly discards intensity gradient information, making the estimated D_B sensitive to the choice of t^*.

To avoid scanning all ε^2 pixels per cell, the optimised formulation pre-computes the *integral image* $P[x][y] = \sum_{i \le x, j \le y} I[i][j]$ using the recurrence (CROW, 1984)

$$P[x][y] = I[x][y] + P[x-1][y] + P[x][y-1] - P[x-1][y-1]. \tag{5}$$

Given P, the foreground pixel count within any rectangular cell is retrieved in $\mathcal{O}(1)$ as

$$\begin{aligned}\sigma(x_1, y_1, x_2, y_2) = {} & P[x_2][y_2] - P[x_1-1][y_2] \\ & - P[x_2][y_1-1] + P[x_1-1][y_1-1],\end{aligned} \tag{6}$$

with occupancy established by $\sigma > 0$. The integral image is built once in $\Theta(N)$ time and space, and shared across all scales in $\mathcal{S}$.

2.3 Gliding Box Method

The Gliding Box (GB) method (ALLAIN; CLOITRE, 1991) replaces the non-overlapping partition of BC with a *dense sliding window*: a box of side L is placed at every valid pixel position (x, y), and the *box mass*

$$m_{x,y}(L) = \sum_{i=0}^{L-1} \sum_{j=0}^{L-1} I[x+i][y+j] \tag{7}$$

is recorded for each of the $T(L) = (W - L + 1)(H - L + 1)$ valid placements. The resulting *mass-frequency distribution* $n(m, L)$ counts how many placements yield mass m, giving the normalised probability $P(m, L) = n(m, L)/T(L)$.

The fractal dimension is estimated from the mean box mass

$$\mu(L) = \sum_{m=0}^{L^2} m\, P(m, L), \tag{8}$$

by substituting $\mu(L)$ for $M(\varepsilon)$ in Eq. (2) and fitting the slope over $\mathcal{S}$ (PLOTNICK *et al.*, 1996). A complementary descriptor, *lacunarity*, is obtained from the first two moments of the distribution:

$$\Lambda(L) = \frac{\mu^{(2)}(L) - [\mu(L)]^2}{[\mu(L)]^2}, \tag{9}$$

where $\mu^{(2)}(L) = \sum_m m^2 P(m, L)$. Lacunarity quantifies translational heterogeneity: a homogeneous set yields $\Lambda \approx 0$, while a clustered or gapped structure yields large Λ (ALLAIN; CLOITRE, 1991; TOLLE; MCJUNKIN; GORSICH, 2008). The joint use of D_B and Λ has been shown to improve tissue discriminability in histological image classification (ROBERTO *et al.*, 2021b; BORGUE *et al.*, 2025b), since two textures may share a similar D_B yet differ in their spatial mass distribution.

The same integral-image structure of Eq. (5) applies to GB: pre-computing P reduces each mass evaluation from $\Theta(L^2)$ to $\mathcal{O}(1)$ via Eq. (6), bringing the per-scale cost from $\Theta(NL^2)$ to $\Theta(N)$ (CROW, 1984). This optimisation is more impactful for GB than for BC because GB evaluates $\Theta(N)$ overlapping positions per scale rather than $\Theta(N/L^2)$ disjoint cells, making the naive cost grow as L^2 while the optimised cost remains constant.

2.4 Differential Box-Counting Method

The Differential Box-Counting (DBC) method, proposed by Sarkar and Chaudhuri (SARKAR; CHAUDHURI, 1994), extends BC to grayscale images by treating the intensity surface as a three-dimensional relief. Let I be a grayscale image of $N = W \times H$ pixels with intensities in $\{0, \ldots, G-1\}$. For a given scale ε, the image plane is partitioned into $\lceil W/\varepsilon \rceil \times \lceil H/\varepsilon \rceil$ non-overlapping cells as in BC, but each cell is now extended vertically: the intensity axis is divided into bands of height $\delta = \lceil G/\lceil W/\varepsilon \rceil \rceil$, and the number of boxes required to cover the local intensity range within cell (i, j) is

$$n_{ij}(\varepsilon) = \left\lceil \frac{I_{\max}^{ij} - I_{\min}^{ij}}{\delta} \right\rceil + 1, \tag{10}$$

where $I_{\max}^{ij}$ and $I_{\min}^{ij}$ are the maximum and minimum intensities within the cell. The total count $N(\varepsilon) = \sum_{i,j} n_{ij}(\varepsilon)$ is substituted into Eq. (2) to estimate D_B via regression over $\mathcal{S}$.

By retaining gradient information through the intensity range $I_{\max}^{ij} - I_{\min}^{ij}$, DBC avoids the threshold sensitivity inherent to BC, at the cost of processing a richer input representation. The naive formulation scans all ε^2 pixels per cell to compute the local extrema, yielding a per-scale cost of $\Theta(N)$ with a constant proportional to ε^2. The optimised formulation replaces this scan with two separate *2D range-minimum and range-maximum structures* built in $\Theta(N \log N)$ time and $\Theta(N \log N)$ space via sparse tables, after which each $(I_{\max}^{ij}, I_{\min}^{ij})$ query is answered in $\mathcal{O}(1)$, reducing the per-scale traversal to $\Theta(N/\varepsilon^2)$ with a unit constant — the same asymptotic profile as the optimised BC applied to a grayscale input.

2.5 Blanket Method

The Blanket Method (BM), introduced by Peleg et al. (PELEG *et al.*, 1984), estimates the fractal dimension of the intensity surface by measuring how its apparent area grows as a morphological cover is progressively expanded around it at each scale. Let $u_\varepsilon(x, y)$ and $b_\varepsilon(x, y)$ denote the upper and lower *blankets* at dilation level ε, defined by the recurrences

$$u_\varepsilon(x, y) = \max\Big(u_{\varepsilon-1}(x, y) + 1, \max_{(s,t)\in\mathcal{N}} u_{\varepsilon-1}(s, t)\Big), \tag{11}$$

$$b_\varepsilon(x, y) = \min\Big(b_{\varepsilon-1}(x, y) - 1, \min_{(s,t)\in\mathcal{N}} b_{\varepsilon-1}(s, t)\Big), \tag{12}$$

with $u_0 = b_0 = I$ and $\mathcal{N}$ denoting the 4-connected neighbourhood of (x, y). The *blanket volume* at scale ε is

$$V(\varepsilon) = \sum_{x,y} \big[u_\varepsilon(x, y) - b_\varepsilon(x, y)\big], \tag{13}$$

and the scale-dependent area measure is obtained as $A(\varepsilon) = \big[V(\varepsilon) - V(\varepsilon-1)\big]/2$. Substituting $A(\varepsilon)$ for $M(\varepsilon)$ in Eq. (2) and fitting the slope over $\mathcal{S}$ yields the fractal dimension estimate.

The naive formulation evaluates each recurrence step by scanning the K-neighbourhood of every pixel, incurring $\Theta(NK^2)$ per dilation level and $\Theta(NK^2\varepsilon_{\max})$ in total. The optimised formulation decomposes the 2D neighbourhood maximum and minimum into two successive 1D sliding-window operations — one horizontal, one vertical — each solved in $\Theta(N)$ using a *monotonic deque* (double-ended queue) per row or column (LEMIRE, 2006). This separability reduces the per-level cost to $\Theta(N)$ regardless of neighbourhood size, and the total cost to $\Theta(N\varepsilon_{\max})$. Because BM operates directly on the intensity surface without binarisation, it retains the full gradient information of the image, making it the most information-preserving of the four methods at the price of higher computational demand.

Algorithm 1 Naive Classical Box-Counting

Require: Binary image $I[0 \dots W-1][0 \dots H-1]$, scale set $\mathcal{S}$
Ensure: Pairs $\{(\varepsilon,\, N(\varepsilon)) : \varepsilon \in \mathcal{S}\}$

```
Binarise I via Otsu's threshold t* (OTSU, 1979)
for each ε ∈ S do
    count ← 0
    for b_x ← 0 to ⌈W/ε⌉ − 1 do
        for b_y ← 0 to ⌈H/ε⌉ − 1 do
            occupied ← FALSE
            for dx ← 0 to ε − 1 do
                for dy ← 0 to ε − 1 do
                    px ← b_x · ε + dx;  py ← b_y · ε + dy
                    if px < W ∧ py < H ∧ I[px][py] = 1 then
                        occupied ← TRUE;  break
                    end if
                end for
            end for
            if occupied then
                count ← count + 1
            end if
        end for
    end for
    yield (ε, count)
end for
Estimate D_B by least-squares regression on {(log ε_i, log N(ε_i))}
```

Algorithm 2 Optimised Box-Counting (Integral Image)

Require: Binary image $I[0 \dots W-1][0 \dots H-1]$, scale set $\mathcal{S}$
Ensure: Pairs $\{(\varepsilon,\, N(\varepsilon)) : \varepsilon \in \mathcal{S}\}$

```
Binarise I via Otsu's threshold t* (OTSU, 1979)
Build integral image P via Eq. (5)   ▷ Θ(N) time and space (CROW, 1984)
for each ε ∈ S do
    count ← 0
    for b_x ← 0 to ⌈W/ε⌉ − 1 do
        for b_y ← 0 to ⌈H/ε⌉ − 1 do
            x_1 ← b_x · ε;  y_1 ← b_y · ε
            x_2 ← min(x_1+ε−1, W−1);  y_2 ← min(y_1+ε−1, H−1)
            if σ(x_1, y_1, x_2, y_2) > 0 then   ▷ Eq. (6), O(1)
                count ← count + 1
            end if
        end for
    end for
    yield (ε, count)
end for
Estimate D_B by least-squares regression on {(log ε_i, log N(ε_i))}
```

3 Algorithm Project and Analysis

This section presents the algorithmic design and formal asymptotic analysis of the four fractal dimension estimators under study. For each method, a naive brute-force formulation is first described through pseudocode and characterised in terms of its time and auxiliary space complexity; an optimised counterpart is then derived by identifying the computational bottleneck and replacing it with a structurally superior data-structure choice. Throughout this section, $N = W \times H$ denotes the total pixel count of the input image, $m = |\mathcal{S}|$ the number of scales in the evaluation set $\mathcal{S}$, and ε (or L, used interchangeably for the GB) the current box side length. Asymptotic bounds are stated in $\mathcal{O}$, Θ, and Ω notation following the Bachmann–Landau conventions; Θ is used whenever both upper and lower bounds coincide, and $\mathcal{O}$ when only an upper bound is asserted.

3.1 Classical Box-Counting (BC)

3.1.1 Naive Formulation

The naive BC algorithm directly implements Eq. (4) by iterating over every grid cell and scanning all pixels within it to test for occupancy. For each scale $\varepsilon \in \mathcal{S}$, the image domain is partitioned into $\lceil W/\varepsilon \rceil \times \lceil H/\varepsilon \rceil$ non-overlapping cells; within each cell, pixels are visited one by one until a foreground pixel is found or the cell is exhausted (MANDELBROT, 1982; LIEBOVITCH; TOTH, 1989). Algorithm 1 details this procedure.

For a fixed ε, the grid contains $\lceil W/\varepsilon \rceil \cdot \lceil H/\varepsilon \rceil = \Theta(N/\varepsilon^2)$ cells, each requiring at most ε^2 pixel inspections in the worst case (fully foreground cell) and as few as one in the best case (first pixel is foreground). The worst-case per-scale cost is therefore

$$T_{\text{BC,naive}}(\varepsilon) = \Theta\left(\frac{N}{\varepsilon^2}\right) \cdot \mathcal{O}(\varepsilon^2) = \mathcal{O}(N). \tag{14}$$

The lower bound is $\Omega(N/\varepsilon^2)$, since every cell must be visited regardless of its content. Beyond the read-only input buffer of size $\Theta(N)$, the naive formulation requires only a constant number of scalar accumulators, yielding $\Theta(1)$ auxiliary space. Summing over all m scales and including the $\Theta(N)$ binarisation pass:

$$T_{\text{BC,naive}}(N, m) = \Theta(N) + \sum_{\varepsilon \in \mathcal{S}} \mathcal{O}(N) = \mathcal{O}(mN). \tag{15}$$

3.1.2 Optimised Formulation

The inner double loop of Algorithm 1 is the computational bottleneck: it re-inspects up to ε^2 pixels per cell at every scale. This bottleneck is eliminated by pre-computing the two-dimensional prefix sum (integral image) P defined in Eq. (5), which allows any rectangular region sum to be answered in $\mathcal{O}(1)$ using Eq. (6) (CROW, 1984). A cell is occupied if and only if $\sigma > 0$, replacing the entire inner scan with a single arithmetic expression. Algorithm 2 presents the optimised procedure.

The integral image is constructed once in $\Theta(N)$ time and shared across all scales. For each $\varepsilon \in \mathcal{S}$, the outer cell loops visit exactly $\Theta(N/\varepsilon^2)$ cells, each processed in $\mathcal{O}(1)$:

$$T_{\text{BC,opt}}(N, m) = \Theta(N) + \sum_{\varepsilon \in \mathcal{S}} \Theta\left(\frac{N}{\varepsilon^2}\right). \tag{16}$$

Algorithm 3 Naive Gliding Box

Require: Binary image $I[0 \ldots W{-}1][0 \ldots H{-}1]$, scale set $\mathcal{S}$
Ensure: Pairs $\{(L,\, \mu(L),\, \Lambda(L)) : L \in \mathcal{S}\}$
Binarise I via Otsu's threshold t^* (OTSU, 1979)
for each $L \in \mathcal{S}$ **do**
    $H[0 \ldots L^2] \leftarrow 0$ ▷ mass-frequency histogram
    for $x \leftarrow 0$ **to** $W - L$ **do**
        for $y \leftarrow 0$ **to** $H - L$ **do**
            $mass \leftarrow 0$
            for $dx \leftarrow 0$ **to** $L - 1$ **do**
                for $dy \leftarrow 0$ **to** $L - 1$ **do**
                    $mass \leftarrow mass + I[x{+}dx][y{+}dy]$
                end for
            end for
            $H[mass] \leftarrow H[mass] + 1$
        end for
    end for
    $T \leftarrow (W{-}L{+}1)(H{-}L{+}1)$
    Compute $\mu(L)$ and $\mu^{(2)}(L)$ from H and T via Eqs. (8)–(**??**)
    Compute $\Lambda(L)$ via Eq. (9)
    yield $(L,\, \mu(L),\, \Lambda(L))$
end for
Estimate D_B by least-squares regression on $\{(\log L_i,\, \log \mu(L_i))\}$ (PLOTNICK *et al.*, 1996)

For the typical scale set $\mathcal{S} = \{2^k : k = 1, \ldots, \lfloor \log_2 \min(W, H) \rfloor\}$, the sum $\sum_\varepsilon N/\varepsilon^2$ is dominated by the smallest scale ($\varepsilon = 2$), giving $\Theta(N/4)$. The total cost is therefore $\Theta(N)$, a strict improvement over the $\mathcal{O}(mN)$ naive bound of Eq. (15). The integral image P requires $\Theta(N)$ auxiliary memory as the sole structure that grows with the input, so the total auxiliary space of the optimised formulation is $\Theta(N)$ — a constant-factor overhead relative to the input. Table 1 consolidates these bounds for both formulations.

Tabela 1: Asymptotic complexity of Classical Box-Counting.

Formulation	**Time**	**Aux. Space**
Naive	$\mathcal{O}(mN)$	$\Theta(1)$
Optimised	$\Theta(N)$	$\Theta(N)$

3.2 Gliding Box (GB)

3.2.1 Naive Formulation

The Gliding Box algorithm, introduced by Allain and Cloitre (ALLAIN; CLOITRE, 1991), replaces the disjoint grid of BC with a dense sliding window. For each scale $L \in \mathcal{S}$, the box is positioned at every valid pixel (x, y), and the box mass $m_{x,y}(L)$ of Eq. (7) is computed by summing all L^2 pixels within the window. The resulting mass-frequency histogram $n(m, L)$ is then normalised to yield $P(m, L)$, from which the mean mass $\mu(L)$ of Eq. (8) and the lacunarity $\Lambda(L)$ of Eq. (9) are computed (PLOTNICK *et al.*, 1996; TOLLE; MCJUNKIN; GORSICH, 2008). Algorithm 3 details the naive procedure.

Algorithm 4 Optimised Gliding Box (Integral Image)

Require: Binary image $I[0 \ldots W{-}1][0 \ldots H{-}1]$, scale set $\mathcal{S}$, integral image P (pre-computed, shared with BC)
Ensure: Pairs $\{(L,\, \mu(L),\, \Lambda(L)) : L \in \mathcal{S}\}$
for each $L \in \mathcal{S}$ **do**
    $H[0 \ldots L^2] \leftarrow 0$
    for $x \leftarrow 0$ **to** $W - L$ **do**
        for $y \leftarrow 0$ **to** $H - L$ **do**
            $mass \leftarrow \sigma(x,\, y,\, x{+}L{-}1,\, y{+}L{-}1)$ ▷ Eq. (6), $\mathcal{O}(1)$ (CROW, 1984)
            $H[mass] \leftarrow H[mass] + 1$
        end for
    end for
    $T \leftarrow (W{-}L{+}1)(H{-}L{+}1)$
    Compute $\mu(L)$ and $\mu^{(2)}(L)$ from H and T
    Compute $\Lambda(L)$ via Eq. (9)
    yield $(L,\, \mu(L),\, \Lambda(L))$
end for
Estimate D_B by least-squares regression on $\{(\log L_i,\, \log \mu(L_i))\}$ (PLOTNICK *et al.*, 1996)

For a fixed scale L, there are $T(L) = (W - L + 1)(H - L + 1) = \Theta(N)$ valid window positions, each requiring a full $L \times L$ pixel scan costing $\Theta(L^2)$, so the per-scale cost is

$$T_{\mathrm{GB,naive}}(L) = \Theta(N) \cdot \Theta(L^2) = \Theta(NL^2). \tag{17}$$

The mass-frequency histogram requires $\Theta(L^2)$ entries at scale L, giving $\mathcal{O}(L^2_{\max})$ auxiliary space — negligible relative to the input when $L_{\max} \ll \sqrt{N}$, but potentially significant for coarse scales. Summing over all m scales and including the binarisation pass:

$$T_{\mathrm{GB,naive}}(N, m) = \Theta(N) + \sum_{L \in \mathcal{S}} \Theta(NL^2) = \Theta\left(N \sum_{L \in \mathcal{S}} L^2\right). \tag{18}$$

For a scale set whose largest element is $L_{\max}$, the sum $\sum L^2$ is $\Theta(mL^2_{\max})$ in the worst case, making the naive GB substantially more expensive than the naive BC for the same scale set. The dependence on L^2 per position constitutes the algorithmic bottleneck identified for optimisation.

3.2.2 Optimised Formulation

The $\Theta(L^2)$ per-position cost of the naive GB arises entirely from the inner double loop that recomputes the box sum from scratch at each window position. This is identical in structure to the inner scan of naive BC, and is eliminated by the same integral-image mechanism: pre-computing P via Eq. (5) and querying σ via Eq. (6) reduces each mass evaluation to $\mathcal{O}(1)$ (CROW, 1984). Since BC and GB share the same input image, the integral image constructed for BC can be reused directly for GB without additional memory allocation, provided the two methods are executed within the same pipeline. Algorithm 4 presents the optimised procedure.

With $\mathcal{O}(1)$ mass queries, the per-scale cost collapses to the number of window positions:

$$T_{\mathrm{GB,opt}}(L) = \Theta(N) \cdot \mathcal{O}(1) = \Theta(N). \tag{19}$$

Including the single shared $\Theta(N)$ prefix-sum construction and the binarisation pass, the total cost becomes

$$T_{\mathrm{GB,opt}}(N, m) = \Theta(N) + \sum_{L \in \mathcal{S}} \Theta(N) = \Theta(mN), \tag{20}$$

a factor of $\Theta(L^2)$ improvement over Eq. (17) at each scale — a gain that grows quadratically with the box side and is therefore most significant at coarse scales. The integral image P is shared with BC and constitutes the sole $\Theta(N)$ auxiliary structure; the per-scale histogram H requires $\Theta(L^2)$ entries, which is typically negligible, keeping the total auxiliary space at $\Theta(N)$. Both optimised formulations therefore share the same $\Theta(N)$ integral image, and their key asymptotic difference lies in the number of positions evaluated per scale: BC visits $\Theta(N/\varepsilon^2)$ disjoint cells, whereas GB visits $\Theta(N)$ overlapping positions regardless of L, incurring a per-scale cost that is a factor of $\Theta(L^2)$ greater than that of optimised BC at the same scale. This overhead is the algorithmic price of the richer mass-frequency distribution that GB produces relative to the binary occupancy count of BC, and which enables the joint estimation of D_B and Λ used in recent histological classification work (ROBERTO *et al.*, 2021b; BORGUE *et al.*, 2025b). Table 2 consolidates the complexity bounds for both GB formulations, and Table 3 contrasts the two methods side by side.

Tabela 2: Asymptotic complexity of the Gliding Box method.

Formulation	**Time**	**Aux. Space**
Naive	$\Theta\big(N \sum_L L^2\big)$	$\mathcal{O}(L^2_{\max})$
Optimised	$\Theta(mN)$	$\Theta(N)$

Tabela 3: BC vs. GB optimised formulations (per scale).

Method	**Positions/scale**	**Cost/position**	**Per-scale cost**
BC (opt.)	$\Theta(N/\varepsilon^2)$	$\mathcal{O}(1)$	$\Theta(N/\varepsilon^2)$
GB (opt.)	$\Theta(N)$	$\mathcal{O}(1)$	$\Theta(N)$

3.3 Differential Box-Counting (DBC)

3.3.1 Naive Formulation

The Differential Box-Counting method, proposed by Sarkar and Chaudhuri (SARKAR; CHAUDHURI, 1994), extends the BC framework to grayscale images by treating the intensity function $z = I(x, y)$ as a three-dimensional surface and counting the number of boxes in the extended (x, y, z) space required to cover the local intensity relief within each spatial cell. Let I be a grayscale image of $N = W \times H$ pixels with intensities in $\{0, \ldots, G-1\}$. For a given scale ε, the image plane is partitioned into $\lceil W/\varepsilon \rceil \times \lceil H/\varepsilon \rceil$ non-overlapping cells as in BC; the intensity axis is simultaneously divided into bands of uniform height

Algorithm 5 Naive Differential Box-Counting

Require: Grayscale image $I[0 \ldots W-1][0 \ldots H-1]$, intensity range G, scale set $\mathcal{S}$
Ensure: Pairs $\{(\varepsilon,\, N(\varepsilon)) : \varepsilon \in \mathcal{S}\}$

for each $\varepsilon \in \mathcal{S}$ **do**
  $\delta \leftarrow \lceil G \,/\, \lceil W/\varepsilon \rceil \rceil$
  $total \leftarrow 0$
  for $b_x \leftarrow 0$ **to** $\lceil W/\varepsilon \rceil - 1$ **do**
    for $b_y \leftarrow 0$ **to** $\lceil H/\varepsilon \rceil - 1$ **do**
      $I_{\max} \leftarrow -\infty$; $I_{\min} \leftarrow +\infty$
      for $dx \leftarrow 0$ **to** $\varepsilon - 1$ **do**
        for $dy \leftarrow 0$ **to** $\varepsilon - 1$ **do**
          $px \leftarrow b_x \cdot \varepsilon + dx$; $py \leftarrow b_y \cdot \varepsilon + dy$
          if $px < W \ \wedge \ py < H$ **then**
            $I_{\max} \leftarrow \max(I_{\max},\, I[px][py])$
            $I_{\min} \leftarrow \min(I_{\min},\, I[px][py])$
          end if
        end for
      end for
      $total \leftarrow total + \lceil (I_{\max} - I_{\min})/\delta \rceil + 1$
    end for
  end for
  yield $(\varepsilon,\, total)$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i,\, \log N(\varepsilon_i))\}$

$$\delta(\varepsilon) = \left\lceil \frac{G}{\lceil W/\varepsilon \rceil} \right\rceil, \tag{21}$$

so that the number of intensity bands scales proportionally with the spatial resolution. Within each cell (i, j), the number of boxes required to cover the intensity range is

$$n_{ij}(\varepsilon) = \left\lceil \frac{I^{ij}_{\max} - I^{ij}_{\min}}{\delta(\varepsilon)} \right\rceil + 1, \tag{22}$$

where $I^{ij}_{\max}$ and $I^{ij}_{\min}$ are the maximum and minimum pixel intensities within cell (i, j). The total count $N(\varepsilon) = \sum_{i,j} n_{ij}(\varepsilon)$ is then substituted into the log-log regression of Eq. (2) to estimate D_B (SARKAR; CHAUDHURI, 1994; FALCONER, 2003). By retaining the intensity gradient $I^{ij}_{\max} - I^{ij}_{\min}$ as the primary signal, DBC avoids the threshold sensitivity inherent to BC while operating directly on the grayscale representation (LIU *et al.*, 2014). Algorithm 5 details the naive procedure.

The structure of the naive DBC is isomorphic to that of naive BC: for each scale ε, the $\Theta(N/\varepsilon^2)$ cells are each traversed by an inner $\varepsilon \times \varepsilon$ loop that now computes both a running maximum and a running minimum rather than a single occupancy flag. Each inner loop iteration performs $\mathcal{O}(1)$ work, so the per-scale cost is

$$T_{\mathrm{DBC,naive}}(\varepsilon) = \Theta\left(\frac{N}{\varepsilon^2}\right) \cdot \Theta(\varepsilon^2) = \Theta(N), \tag{23}$$

and the total cost over m scales is

$$T_{\mathrm{DBC,naive}}(N, m) = \sum_{\varepsilon \in \mathcal{S}} \Theta(N) = \Theta(mN). \tag{24}$$

Algorithm 6 Optimised Differential Box-Counting (Sparse Tables)

Require: Grayscale image $I[0\ldots W-1][0\ldots H-1]$, intensity range G, scale set $\mathcal{S}$
Ensure: Pairs $\{(\varepsilon,\ N(\varepsilon)) : \varepsilon \in \mathcal{S}\}$

Build 2D sparse table $\mathcal{T}_{\max}$ for range maximum ▷ $\Theta(N \log N)$ time and space (LIU *et al.*, 2014)
Build 2D sparse table $\mathcal{T}_{\min}$ for range minimum ▷ $\Theta(N \log N)$ time and space
for each $\varepsilon \in \mathcal{S}$ **do**
  $\delta \leftarrow \lceil G \,/\, \lceil W/\varepsilon \rceil \rceil$
  $total \leftarrow 0$
  for $b_x \leftarrow 0$ **to** $\lceil W/\varepsilon \rceil - 1$ **do**
    for $b_y \leftarrow 0$ **to** $\lceil H/\varepsilon \rceil - 1$ **do**
      $x_1 \leftarrow b_x \cdot \varepsilon$; $y_1 \leftarrow b_y \cdot \varepsilon$
      $x_2 \leftarrow \min(x_1+\varepsilon-1,\ W-1)$; $y_2 \leftarrow \min(y_1+\varepsilon-1,\ H-1)$
      $I_{\max} \leftarrow \mathcal{T}_{\max}.\textsc{Query}(x_1, y_1, x_2, y_2)$ ▷ $\mathcal{O}(1)$
      $I_{\min} \leftarrow \mathcal{T}_{\min}.\textsc{Query}(x_1, y_1, x_2, y_2)$ ▷ $\mathcal{O}(1)$
      $total \leftarrow total + \lceil (I_{\max} - I_{\min})/\delta \rceil + 1$
    end for
  end for
  yield $(\varepsilon,\ total)$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i,\ \log N(\varepsilon_i))\}$

Although the asymptotic bound matches that of naive BC, DBC carries a larger constant factor per cell visit, since two comparisons are performed per pixel instead of one, and no early-exit is possible — the entire cell must be scanned to determine $I^{ij}_{\max}$ and $I^{ij}_{\min}$. Auxiliary space is $\Theta(1)$ beyond the read-only input buffer, identical to the naive BC case.

3.3.2 Optimised Formulation

The computational bottleneck of the naive DBC is the $\Theta(\varepsilon^2)$ inner scan required to compute the local range $(I^{ij}_{\max},\ I^{ij}_{\min})$ at every cell and every scale. This bottleneck is eliminated by pre-computing two 2D range structures — one for maxima and one for minima — that answer arbitrary rectangular range queries in $\mathcal{O}(1)$ after an offline preprocessing phase. The standard construction uses *sparse tables* (LIU *et al.*, 2014): for each row, a 1D sparse table is built in $\mathcal{O}(W \log W)$ time; a second-level sparse table over the row results is then built in $\mathcal{O}(H \log H)$ time, yielding a total preprocessing cost of $\Theta(N \log N)$ time and $\Theta(N \log N)$ auxiliary space. Once built, any $(I^{ij}_{\max},\ I^{ij}_{\min})$ query over a cell of arbitrary side is answered in $\mathcal{O}(1)$. Algorithm 6 presents the optimised procedure.

With $\mathcal{O}(1)$ range queries, the per-scale traversal visits $\Theta(N/\varepsilon^2)$ cells at unit cost each, mirroring the optimised BC profile. The total cost becomes

$$T_{\mathrm{DBC,opt}}(N, m) = \Theta(N \log N) + \sum_{\varepsilon \in \mathcal{S}} \Theta\left(\frac{N}{\varepsilon^2}\right) = \Theta(N \log N), \tag{25}$$

where the preprocessing term $\Theta(N \log N)$ dominates for any fixed scale set. The auxiliary space is $\Theta(N \log N)$ for the two sparse tables — a higher memory footprint than the $\Theta(N)$ integral image of BC and GB, which constitutes the primary space trade-off of the DBC optimisation. Table 4 summarises the complexity bounds for both DBC formulations.

Tabela 4: Asymptotic complexity of Differential Box-Counting.

Formulation	Time	Aux. Space
Naive	$\Theta(mN)$	$\Theta(1)$
Optimised	$\Theta(N \log N)$	$\Theta(N \log N)$

3.4 Blanket Method (BM)

3.4.1 Naive Formulation

The Blanket Method, introduced by Peleg et al. (PELEG *et al.*, 1984), estimates the fractal dimension of the intensity surface $z = I(x, y)$ by measuring the growth rate of the surface's apparent area as a morphological cover is progressively expanded around it. Two complementary surfaces — an *upper blanket* $u_\varepsilon(x, y)$ and a *lower blanket* $b_\varepsilon(x, y)$ — are constructed by iterated dilation and erosion, respectively, starting from the initial condition $u_0 = b_0 = I$ and advancing through the recurrences

$$u_\varepsilon(x, y) = \max\left\{ u_{\varepsilon-1}(x, y) + 1,\ \max_{(s,t) \in \mathcal{N}(x,y)} u_{\varepsilon-1}(s, t) \right\}, \tag{26}$$

$$b_\varepsilon(x, y) = \min\left\{ b_{\varepsilon-1}(x, y) - 1,\ \min_{(s,t) \in \mathcal{N}(x,y)} b_{\varepsilon-1}(s, t) \right\}, \tag{27}$$

where $\mathcal{N}(x, y)$ denotes the 4-connected neighbourhood of pixel (x, y). The blanket volume at dilation level ε is

$$V(\varepsilon) = \sum_{x,y} \big[u_\varepsilon(x, y) - b_\varepsilon(x, y) \big], \tag{28}$$

and the scale-dependent area measure is obtained as $A(\varepsilon) = [V(\varepsilon) - V(\varepsilon - 1)]/2$. Substituting $A(\varepsilon)$ for $M(\varepsilon)$ in Eq. (2) and fitting the slope over a set of dilation levels $\mathcal{S} = \{1, 2, \ldots, \varepsilon_{\max}\}$ yields the fractal dimension estimate (PELEG *et al.*, 1984; FALCONER, 2003). Because the blankets are built directly from the grayscale intensity surface without binarisation, BM retains the full gradient information of the image, making it the most information-preserving of the four methods evaluated in this work. Algorithm 7 details the naive procedure.

Each dilation step iterates over all N pixels; for each pixel, the 4-connected neighbourhood $\mathcal{N}$ is examined in $\mathcal{O}(1)$ (four fixed neighbours), giving a per-level cost of $\Theta(N)$. Since $\varepsilon_{\max}$ levels are processed sequentially, the total cost of the naive BM is

$$T_{\mathrm{BM,naive}}(N, \varepsilon_{\max}) = \Theta(N \cdot \varepsilon_{\max}). \tag{29}$$

Two auxiliary arrays of size N are maintained at all times (the current and previous blanket pair), giving $\Theta(N)$ auxiliary space. Although the per-level constant factor

Algorithm 7 Naive Blanket Method

Require: Grayscale image $I[0 \dots W{-}1][0 \dots H{-}1]$, maximum dilation level $\varepsilon_{\max}$
Ensure: Pairs $\{(\varepsilon,\ A(\varepsilon)) : \varepsilon = 1, \dots, \varepsilon_{\max}\}$

$u \leftarrow I;\ \ b \leftarrow I$ ▷ initialise both blankets
$V_{\text{prev}} \leftarrow \sum_{x,y}[u[x][y] - b[x][y]]$
for $\varepsilon \leftarrow 1$ **to** $\varepsilon_{\max}$ **do**
  $u' \leftarrow$ new array of size $W \times H$
  $b' \leftarrow$ new array of size $W \times H$
  for $x \leftarrow 0$ **to** $W - 1$ **do**
    for $y \leftarrow 0$ **to** $H - 1$ **do**
      $u'[x][y] \leftarrow u[x][y] + 1$
      $b'[x][y] \leftarrow b[x][y] - 1$
      for each $(s, t) \in \mathcal{N}(x, y)$ **do**
        $u'[x][y] \leftarrow \max(u'[x][y],\ u[s][t])$
        $b'[x][y] \leftarrow \min(b'[x][y],\ b[s][t])$
      end for
    end for
  end for
  $u \leftarrow u';\ \ b \leftarrow b'$
  $V \leftarrow \sum_{x,y}[u[x][y] - b[x][y]]$
  yield $\big(\varepsilon,\ (V - V_{\text{prev}})/2\big)$
  $V_{\text{prev}} \leftarrow V$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i,\ \log A(\varepsilon_i))\}$

is small — only four neighbour comparisons per pixel — the sequential dependence between levels (each recurrence depends on the previous blanket) prevents parallelism across levels and makes the $\varepsilon_{\max}$ factor unavoidable in the naive formulation.

3.4.2 Optimised Formulation

A critical observation regarding Eqs. (26)–(27) is that, at each dilation level ε, the update for u_ε requires the maximum of $u_{\varepsilon-1}$ over a $(2\varepsilon+1) \times (2\varepsilon+1)$ neighbourhood — equivalently, a 2D sliding-window maximum with window side $K = 2\varepsilon + 1$. The naive implementation of this 2D window operation visits all $K^2 = \Theta(\varepsilon^2)$ neighbours per pixel, raising the per-level cost to $\Theta(N\varepsilon^2)$ and the total cost to $\Theta(N\varepsilon^3_{\max})$ when generalised to variable-radius dilation. This ε^2 factor is eliminated by exploiting the *separability* of the rectangular maximum and minimum filters: a 2D sliding-window extremum over a $K \times K$ window can be decomposed into two successive 1D sliding-window passes — one horizontal of length K applied to each row, followed by one vertical of length K applied to each column of the intermediate result (LEMIRE, 2006). Each 1D pass is solved in $\Theta(N)$ using a *monotonic deque* (double-ended queue) that maintains a decreasing (for maximum) or increasing (for minimum) sequence of candidate extrema over the active window, processing each pixel with at most two deque operations — one enqueue and at most one dequeue — giving exactly $\Theta(N)$ total operations per pass regardless of K, as established by Lemire (LEMIRE, 2006). The two separable passes together cost $\Theta(N)$ per dilation level, reducing the total BM cost to

Algorithm 8 Optimised Blanket Method (Separable Monotonic Deque)

Require: Grayscale image $I[0 \dots W{-}1][0 \dots H{-}1]$, maximum dilation level $\varepsilon_{\max}$
Ensure: Pairs $\{(\varepsilon,\ A(\varepsilon)) : \varepsilon = 1, \dots, \varepsilon_{\max}\}$

$u \leftarrow I;\ \ b \leftarrow I$
$V_{\text{prev}} \leftarrow \sum_{x,y}[u[x][y] - b[x][y]]$
for $\varepsilon \leftarrow 1$ **to** $\varepsilon_{\max}$ **do**
  $u_{\text{inc}} \leftarrow u + 1$ ▷ element-wise increment, $\Theta(N)$
  $b_{\text{dec}} \leftarrow b - 1$
  $u' \leftarrow \textsc{SlidingMax2D}(u,\ K{=}3)$ ▷ separable deque, $\Theta(N)$ (LEMIRE, 2006)
  $b' \leftarrow \textsc{SlidingMin2D}(b,\ K{=}3)$ ▷ separable deque, $\Theta(N)$
  $u \leftarrow \max(u_{\text{inc}},\ u')$ ▷ element-wise, $\Theta(N)$
  $b \leftarrow \min(b_{\text{dec}},\ b')$
  $V \leftarrow \sum_{x,y}[u[x][y] - b[x][y]]$
  yield $\big(\varepsilon,\ (V - V_{\text{prev}})/2\big)$
  $V_{\text{prev}} \leftarrow V$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i,\ \log A(\varepsilon_i))\}$

$$T_{\text{BM,opt}}(N, \varepsilon_{\max}) = \Theta(N \cdot \varepsilon_{\max}), \tag{30}$$

which matches the naive total of Eq. (29) in asymptotic class but with a strictly smaller constant: the naive approach costs $4N$ operations per level (four fixed-neighbourhood comparisons), while the optimised approach costs $4N$ deque operations per level (two passes of two operations each), so the constant is identical for the 4-connected case. The optimisation becomes decisive when BM is generalised to larger structuring elements or when the dilation is implemented as a single multi-scale pass over the full $K \times K$ window, where the naive cost is $\Theta(NK^2)$ per level versus $\Theta(N)$ for the separable deque formulation. Algorithm 8 presents the optimised procedure.

The auxiliary space requirement of the optimised BM is $\Theta(N)$ — four arrays of size N (two blankets and two intermediate buffers) plus the $\mathcal{O}(K)$ deque storage per row or column pass, which is negligible. Table 5 consolidates the complexity bounds for both BM formulations, and Table 6 provides a consolidated comparison of all four methods.

Tabela 5: Asymptotic complexity of the Blanket Method.

Formulation	**Time**	**Aux. Space**
Naive	$\Theta(N\varepsilon_{\max})$	$\Theta(N)$
Optimised	$\Theta(N\varepsilon_{\max})$	$\Theta(N)$

Tabela 6: Consolidated asymptotic complexity of all four methods (optimised formulations).

Method	**Input**	**Time**	**Aux. Space**
BC (opt.)	Binary	$\Theta(N)$	$\Theta(N)$
GB (opt.)	Binary	$\Theta(mN)$	$\Theta(N)$
DBC (opt.)	Grayscale	$\Theta(N \log N)$	$\Theta(N \log N)$
BM (opt.)	Grayscale	$\Theta(N\varepsilon_{\max})$	$\Theta(N)$

4 Methodology

4.1 Materials

The experimental evaluation is conducted on the UCSB Breast Cancer Histology dataset, provided by the Center for Bio-Image Informatics of the University of California, Santa Barbara (GELASCA *et al.*, 2008). The dataset comprises 58 hematoxylin and eosin (H&E) stained histological images of breast tissue at a fixed resolution of 896×768 pixels, partitioned into two classes: 32 benign and 26 malignant samples. The class imbalance ratio of approximately 1.23:1 is mild enough to not require resampling for the fractal dimension estimation task, since D is computed independently per image and no classifier is trained on the extracted features.

This dataset was selected for three reasons directly relevant to the algorithmic focus of this work. First, its fixed resolution ensures that the pixel count $N = 896 \times 768 = 688{,}128$ is identical across all images, making wall-clock time comparisons across methods and formulations free of resolution confounds. Second, the two well-defined tissue classes — benign and malignant — provide a concrete discriminatory benchmark against which the effectiveness of each algorithm's D estimate can be assessed, following the methodology established in prior fractal-based histological classification work (ROBERTO *et al.*, 2021b; BORGUE *et al.*, 2025b). Third, the dataset is publicly available as a Kaggle dataset (`ucsb-breast-cancer-histology`), enabling direct reproducibility of all experiments within the execution environment described in Section 4.2.

All images are loaded as 8-bit grayscale arrays ($G = 256$) prior to processing. For the binary-domain methods (BC and GB), a binarisation step is applied using Otsu's global threshold (OTSU, 1979) before the fractal dimension estimation loop; this step is timed and reported separately from the algorithmic core, as described in Section 4.2.

4.2 Experimental Environment and Protocol

All experiments are executed on Kaggle Notebooks (⟨https://www.kaggle.com⟩) using a CPU-only session, to ensure that the measured execution times reflect the sequential computational complexity of each algorithm without GPU acceleration. The hardware configuration provided by the Kaggle CPU session consists of an Intel Xeon processor with two virtual cores and approximately 13 GB of available RAM. The software environment is fixed at Python 3.10 with NumPy 1.26, SciPy 1.11, and scikit-image 0.22; the complete `requirements.txt` is included in the *github* repository to ensure reproducibility. [1]

The scale set used across all four methods is fixed as $\mathcal{S} = \{2, 4, 8, 16, 32, 64\}$, comprising $m = 6$ scales spanning approximately 1.5 decades of spatial frequency. This range respects the lower bound imposed by the pixel size and the upper bound of $L_{\max} = 64 \ll \sqrt{N} \approx 830$, ensuring that the covering assumption underlying each algorithm remains valid throughout the scale range. The same $\mathcal{S}$ is used for all methods to guarantee that differences in execution time and R^2 are attributable solely to algorithmic structure rather than to differences in scale coverage.

Each algorithm — in both its naive and optimised formulations — is executed 10 times per image. The first run is discarded as a warm-up to eliminate cold-cache and interpreter overhead effects; the remaining 9 runs are used to compute the mean wall-clock time $\bar{t}$ and standard deviation σ_t. Wall-clock time is measured using Python's `time.perf_counter()`, which provides sub-microsecond resolution on the target platform. Runs exhibiting a coefficient of variation

$$CV = \frac{\sigma_t}{\bar{t}} \times 100\% \tag{31}$$

exceeding 5% are flagged and re-evaluated with additional runs until $CV < 5\%$, following the adaptive stability criterion recommended for wall-clock benchmarking (FLEMMING; WALLACE, 1986). This threshold ensures that the reported means are statistically stable without imposing an unnecessarily large number of repetitions.

Pre-processing time is measured and reported separately from the algorithmic core. Concretely, for BC and GB, the wall-clock time of the Otsu binarisation step is recorded independently of the subsequent coverage computation, allowing the contribution of the pre-processing stage to the total execution budget to be quantified and discussed in Section **??**. For DBC and BM, which operate directly on the grayscale input, no binarisation is performed and pre-processing time is zero by definition.

4.3 Evaluation Metrics

Three orthogonal evaluation axes are considered, each capturing a distinct dimension of the algorithmic trade-off under study.

4.3.1 Execution Time

The primary time metric is the mean wall-clock time $\bar{t}$ (in seconds) averaged over all 58 images and 9 valid runs per image. To assess the agreement between theoretical predictions and empirical scaling behaviour, $\bar{t}$ is plotted as a function of image size N across sub-images of increasing resolution, and the empirical slope of the $\log \bar{t}$ versus $\log N$ curve is compared against the theoretical exponent derived in Section 3. Deviations between empirical and theoretical slopes are attributed to constant factors, memory hierarchy effects, and interpreter overhead, and are discussed qualitatively in Section **??**.

4.3.2 Memory Consumption

Auxiliary memory consumption is measured as the peak heap allocation incurred by each algorithm beyond the read-only input buffer, using Python's `tracemalloc` module. The input buffer of size $\Theta(N)$ bytes is excluded

[1] Source code available at: `https://github.com/gabrielmbrum`.

from the measurement by recording the allocation differential between the start and end of the algorithmic core. This isolates the auxiliary space — the integral image for BC and GB, the sparse tables for DBC, and the blanket arrays for BM — and allows direct comparison against the $\Theta(N)$ and $\Theta(N \log N)$ theoretical bounds derived in Section 3.

4.3.3 Effectiveness

Effectiveness is defined as the capacity of each algorithm's D estimate to statistically discriminate between the benign and malignant tissue classes of the UCSB dataset. Formally, for each method, D is computed for every image and the class-conditional distributions $\{D_i : i \in \text{benign}\}$ and $\{D_j : j \in \text{malignant}\}$ are obtained. The primary effectiveness metric is the inter-class separation

$$\Delta D = \bar{D}_{\text{malignant}} - \bar{D}_{\text{benign}}, \tag{32}$$

where $\bar{D}_{\text{malignant}}$ and $\bar{D}_{\text{benign}}$ denote the respective class means. A larger $|\Delta D|$ indicates that the method assigns meaningfully different fractal dimensions to the two tissue types, reflecting the clinically established observation that malignant tissues exhibit higher surface roughness and structural complexity than benign ones (LOPES; BETROUNI, 2011; RANGAYYAN; NGUYEN, 2007). Statistical significance of ΔD is assessed via Welch's two-sample t-test at significance level $\alpha = 0.05$, which does not assume equal variances across classes and is appropriate given the mild class imbalance of the dataset (WELCH, 1947).

Two complementary effectiveness indicators are also reported. The coefficient of determination R^2 of the log-log regression used to estimate D from each image quantifies the degree to which the object exhibits consistent power-law scaling across the chosen scale set; values close to unity are a necessary condition for interpreting D as a valid fractal descriptor (LOPES; BETROUNI, 2011). Additionally, for the binary methods (BC and GB), the sensitivity of ΔD to the Otsu threshold is evaluated by perturbing t^* by $\pm 10\%$ and measuring the resulting variation in D, providing a quantitative characterisation of the threshold sensitivity trade-off that distinguishes binary from grayscale methods.

5 Results and Discussion

Resultados da pesquisa

Tabela 7: Legenda

col1	bol2	bol3	col4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
resultado1	**resultado1**	**resultado1**	**resultado1**

nota de rodapé

6 Conclusion

Retomada da proposta, resultados e possíveis melhorias/trabalhos futuros.

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